Roll Number: _____

Thapar Institute of Engineering & Technology, Patiala

Department of Computer Science &Engineering

B. E.(COE/CSE) : EST (AUX)	Course Code: UCS510 Course Name: Computer Organization and Architecture
Date: Feb, 27, 2024	Time: 5:30 PM - 8:30 PM
Time: 3 Hours, M. Marks: 100	Faculty: SOL

Note: All questions carry equal marks. Attempt all questions in order. Assume missing data if any.

Q.1.	A. (a) Consider a 4-way set associative mapped cache with block size 4 KB. The size of main memory is 16 GB and there are 10 bits in the tag. Compute: i. Size of cache memory ii. Tag directory size (b) Write the five difference between RISC and CISC architecture. B. (a) Write the difference between memory reference instructions and register reference instructions with one example of both. (b) Differentiate between Interrupted I/O and Programmed I/O.	**10+10**
Q.2	A. Consider a non-pipelined processor with a clock rate of 3.6 gigahertz and average cycles per instruction of 5. The same processor is upgraded to a pipelined processor with six stages, but due to the internal pipeline delay, the clock speed is reduced to 2.5 gigahertz. Assume there are stalls of 1 ns after each stage in the pipeline. Find the speed-up achieved in this pipelined processor. Show each step clearly. B. (a) What are the advantage of addressing mode in instruction format? Write the different addressing modes you studied? **(five only)** Explain each of them with an example. (b) Minimize the following Boolean function using K-map method and write the minimized Boolean expression. $F(A, B, C, D) = \Sigma m(0, 1, 2, 5, 7, 8, 9, 10, 13, 15)$	**10+10**

<table>
<tr><td>Q.3</td><td>A. Consider a processor with 64 registers and an instruction set of size twelve. Each instruction has five distinct fields, namely, opcode, two source register identifiers, one destination register identifier, and a twelve-bit immediate value. Each instruction must be stored in memory in a byte-aligned fashion. If a program has 100 instructions, what would be the amount of memory (in bytes) consumed by the program text?
B. Convert the following numbers in IEEE-754 floating point 32-bit short precision format. (i). $(18.25)_{10}$ (ii). (CA64)H</td><td>10+10</td></tr>
<tr><td>Q.4</td><td>A. Perform the signed multiplication of the following number using Booth's Algorithm with clear steps [12*5] using 5 bits to represent both multiplier and multiplicands.
B. Explain the detailed working of DMA with an interactive diagram.</td><td>10+10</td></tr>
<tr><td>Q.5</td><td>A. Define hazards in the pipeline. What are the different types of hazards you have studied explain each of them with suitable examples and write the solution to resolve each of them?
B. Design an arithmetic circuit with one selection variable S_0 and two 4-bit data inputs A and B. The circuit generates the following four arithmetic operations in conjunction with the input carry C_{in}. Draw the logic diagram for the first two stages.

<table><tr><td></td><td>$C_{in}=0$</td><td>$C_{in}=1$</td></tr><tr><td>$S_0=0$</td><td>A+B</td><td>A+1</td></tr><tr><td>$S_0=1$</td><td>Transfer A</td><td>A-B</td></tr></table></td><td>10+10</td></tr>
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